

Permutation Rounds

Time limit: 1.00 s

Memory limit: 512 MB

There is a sorted array $[1, 2, \dots, n]$ and a permutation $p_1, p_2, \dots, p_n$. On each round, all elements move according to the permutation: the element at position i moves to position p_i . After how many rounds is the array sorted again for the first time?

Input

The first line has an integer n .
The next line contains n integers $p_1, p_2, \dots, p_n$.

Output

Print the number of rounds modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
8 5 3 2 6 4 1 8 7	4

Explanation: The array changes as follows after the rounds:

- Round 1: $[6, 3, 2, 5, 1, 4, 8, 7]$
- Round 2: $[4, 2, 3, 1, 6, 5, 7, 8]$
- Round 3: $[5, 3, 2, 6, 4, 1, 8, 7]$
- Round 4: $[1, 2, 3, 4, 5, 6, 7, 8]$